Off-the-Shelf LLMs as Process Scorers: Training-Free Alternative to PRMs for Mathematical Reasoning

Atoosa Chegini[1], **Soheil Feizi**[1]
[1]Department of Computer Science, University of Maryland
Correspondence: atoocheg@umd.edu

Abstract

Selecting the best response from multiple small-model samples using a stronger scorer is a simple inference-time strategy, but fails in practice because the small model has already committed to diverging reasoning paths that no scorer can correct; PRM guided search avoids this by scoring candidate continuations during generation, but requires a reward model trained with step-level labels.

We propose *Chunk-Level Guided Generation*, a training-free alternative that uses an off-the-shelf large language model as a process scorer: a small model samples k fixed-length candidate chunks at each step, while the larger language model scores the candidates using likelihoods, without generating any text. The selected chunk is committed before the next step, steering generation before errors can propagate.

We instantiate this framework with two selection rules: *Likelihood-Guided Selection (LGS)*, which selects the chunk with the highest length-normalized large-model log-probability, and *Contrastive-Guided Selection (CGS)*, which subtracts the small model's log-probability to favor chunks where the large model's preference diverges from the small model's. We show that scoring variable-length reasoning steps with large-model likelihoods is unreliable due to a systematic length bias that persists even after length normalization, and that fixed-length chunks avoid this confound.

On GSM8K, MATH, Minerva Math, AMC23, and AIME24 with Qwen2.5-1.5B guided by Qwen2.5-32B and Llama-3.2-1B guided by Llama-3.1-70B, CGS outperforms majority voting by up to 28 pp (e.g., 83.9% vs. 56.0% on GSM8K with the Llama pair at k=32) and, under matched guidance budgets, matches or outperforms Qwen2.5-Math-PRM-72B guided search on most benchmarks without any reward-model training (e.g., 39.0% vs. 32.4% on Minerva Math with the Qwen pair at k=32). With Qwen2.5-7B guided by Qwen2.5-72B, CGS reaches 81.8% on MATH and 63.6% on Minerva Math at k=16, surpassing majority voting by 4–6 pp.

Finally, we show that Chunk-Level Guided Generation produces substantially shorter reasoning traces than PRM guided search, with PRM responses up to 80% longer on GSM8K.

1 Introduction

Large language models have made substantial progress on mathematical reasoning, but using a large model for every inference query remains expensive. A practical alternative is to sample multiple solutions from a smaller model and then select the best one. Majority voting (Wang et al., 2023), verifier-based Best-of-N (Cobbe et al., 2021), and self-certainty-based selection (Kang et al., 2025) follow this post-hoc paradigm: the small model first generates complete candidate solutions, and selection happens only after generation has finished.

These post-hoc methods are simple, but they have an important limitation. Small-model samples can diverge early into different incorrect reasoning paths. Once every candidate has already committed to flawed intermediate steps, even a strong scorer can only choose among completed trajectories; it cannot intervene before the error propagates.

Process reward model (PRM) guided search (She et al., 2025; Snell et al., 2024) addresses this timing problem by moving selection inside generation. Instead of scoring only completed responses, it repeatedly samples candidate reasoning steps, scores them with a trained PRM, and appends the highest-scoring step before continuing generation. This generate-score-select loop can guide intermediate reasoning decisions, but it requires a reward model trained for step-level evaluation (Lightman et al., 2023), such as Qwen2.5-Math-PRM-72B (Zhang et al., 2025).

We propose *Chunk-Level Guided Generation*, a training-free alternative that keeps the generation-time selection structure of PRM guided search but

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removes the need for a trained reward model. At each step, a small model samples k fixed-length candidate chunks of length L. A larger language model scores candidate chunks using likelihoods, and the highest-scoring chunk is appended to the context before generation continues. The large model never generates text for the final answer; it only scores candidate continuations produced by the small model, which can be evaluated in parallel.

We study two scoring rules. *Likelihood-Guided Selection (LGS)* selects the chunk with the highest length-normalized large-model log-probability. *Contrastive-Guided Selection (CGS)* subtracts the small model's likelihood from the large model's likelihood, favoring chunks that the larger model finds more plausible than the small model. This emphasizes continuations where the larger model contributes information beyond the small model's own preferences.

A natural question is whether large-model likelihoods are already sufficient for post-hoc selection. We show they are not: Best-of-N with large-model likelihood scoring often underperforms majority voting on MATH and AMC23, and can even degrade as k grows (55.8% $\rightarrow$ 52.6% on MATH with Qwen2.5-1.5B guided by Qwen2.5-32B). The failure is not the scorer — the same likelihood signal drives substantial gains at chunk level. The bottleneck is timing.

A key design choice is to guide generation using fixed-length chunks rather than naturally delimited reasoning steps. We show that large-model likelihoods are biased toward longer reasoning steps even after length normalization. Fixed-length chunks avoid this confound because all candidates at a given decision point have the same length, making their scores directly comparable.

We evaluate *Chunk-Level Guided Generation* on five mathematical reasoning benchmarks: GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al., 2021), Minerva Math (Lewkowycz et al., 2022), AMC23[1], and AIME24[2]. Across Qwen2.5-1.5B guided by Qwen2.5-32B (Team, 2024) and Llama-3.2-1B guided by Llama-3.1-70B (Meta AI, 2024), chunk-level guidance substantially outperforms post-hoc selection methods, including Majority@k (Wang et al., 2023), Self-Certainty and Borda count (Kang et al., 2025), and Best-of-N (Cobbe et al., 2021). Under matched guidance budgets, our *Contrastive-Guided Selection* matches or outperforms PRM guided search using Qwen2.5-Math-PRM-72B (Yang et al., 2024; Zhang et al., 2025) on most benchmarks, without using any reward model.

The gains also persist when the generator is scaled to Qwen2.5-7B (Team, 2024) guided by Qwen2.5-72B. CGS achieves the best average accuracy among methods that do not use a trained reward model, outperforming the strongest post-hoc baseline on average across five benchmarks. It comes within 0.4 pp of the standalone 72B greedy baseline and within 1.5 pp of PRM-72B guided search on average, while requiring no reward-model training.

Finally, we find that these gains do not come from generating longer solutions: CGS produces substantially shorter reasoning traces than PRM guided search, with PRM responses containing up to 80% more tokens on GSM8K.

Our contributions are:

- We show that an off-the-shelf large language model, used only as a likelihood scorer and without any reward-model training, is an effective scorer for generation-time guidance. We instantiate this idea with two selection rules: Likelihood-Guided Selection (ℓ_{π_l}) and Contrastive-Guided Selection ($\ell_{\pi_l} - \ell_{\pi_s}$).

- We show that large-model log-probabilities are unreliable for scoring variable-length reasoning steps due to a systematic length bias that persists even after length normalization, and that fixed-length chunks resolve this.

- Our results show consistent gains over post-hoc selection methods across three model pairs: Qwen2.5-1.5B guided by 32B, Llama-3.2-1B guided by 70B, and Qwen2.5-7B guided by 72B. CGS matches or surpasses PRM-72B guided search in the two smaller-generator settings, and comes within 1.5 pp of PRM-72B on average in the 7B setting, while requiring no reward-model training.

- Chunk-Level Guided Generation produces substantially shorter reasoning traces than PRM guided search, with PRM responses averaging 80% more tokens on GSM8K.

[1] https://huggingface.co/datasets/math-ai/amc23

[2] https://huggingface.co/datasets/HuggingFaceH4/aime_2024

2 Related Work

Sampling and post-hoc selection. Self-consistency (Wang et al., 2023) samples multiple chain-of-thought solutions and takes a majority vote over their final answers. Verifier-based Best-of-N methods (Cobbe et al., 2021) generate complete responses, score them with a verifier, and select the highest-ranked response. Snell et al. (2024) show that Best-of-N becomes increasingly suboptimal on harder problems as test-time compute grows. Kang et al. (2025) propose self-certainty as a training-free confidence signal for Best-of-N selection. These methods operate *post-hoc*: generation is complete before selection, so early reasoning errors can only be avoided if another sampled trajectory corrects them.

Trained scorers and step-level guidance. Process reward models (PRMs) are trained to assign scores to individual steps in a reasoning trajectory, using either human annotations (Lightman et al., 2023; Uesato et al., 2022) or automated rollout- or search-based labels (Wang et al., 2024; Luo et al., 2024). Yuan et al. (2024) reduce this supervision cost by deriving PRM-like signals from outcome-supervised reward models, avoiding explicit step-level labels while still requiring reward-model training. At inference time, She et al. (2025) propose a guided search procedure that generates N candidate continuations for the next reasoning step, scores each with a trained PRM, and appends the best; Khalifa et al. (2023) follow a similar generate-score-select structure but use a discriminator trained to distinguish correct from incorrect reasoning steps. Unlike these approaches, our method requires no reward-model training and uses an off-the-shelf large language model to score fixed-length chunks of L tokens, rather than variable-length reasoning steps; we compare directly against Qwen2.5-Math-PRM-72B (Zhang et al., 2025).

Large-to-small collaborative generation. Recent training-free methods improve small-model reasoning by having a larger model contribute text at inference time. Speculative Thinking (Yang et al., 2025) delegates selected reflective reasoning steps to a larger model; MentorCollab (Wang et al., 2026) injects large-model lookahead segments when a verifier predicts guidance is useful; and Tandem (Fu et al., 2026) primes the small model with compact reasoning insights generated by the large model upfront. In contrast, our large model contributes no generated text: the small model is the sole generator, and the large model only scores fixed-length chunks already sampled from it.

Contrastive and speculative decoding. Contrastive decoding (Li et al., 2023; O'Brien and Lewis, 2023) improves text quality and reasoning by modifying the decoding distribution at each token step of a single generation, upweighting tokens the large model favors over the small model using $\log \pi_{\text{large}}(x) - \log \pi_{\text{small}}(x)$ and suppressing tokens both models agree on; our CGS adapts this signal to chunk-level selection among k small-model candidates rather than token-level steering. Classical speculative decoding (Leviathan et al., 2023; Chen et al., 2023) uses a small draft model to propose tokens that a larger target model verifies in parallel, reducing latency while preserving the target model's output distribution. RSD (Liao et al., 2025) incorporates a process reward model to evaluate draft tokens and dynamically decide whether to invoke the target model, biasing selection toward higher-reward outputs; SPECS (Cemri et al., 2025) generates candidate reasoning traces with a draft model and scores them using both the target model and a reward model to improve accuracy at test time.

3 Method

Let π_s and π_l denote the small and large language models. Given a question q, let h denote the current generation context, initialized as $h = q$. At each iteration, the small model samples k candidate continuations of length L tokens, $c_1, \ldots, c_k$, autoregressively from $\pi_s(\cdot \mid h)$, where L is a hyperparameter controlling chunk length. For a candidate chunk $c = (c^1, \ldots, c^{|c|})$ where $|c| = L$ unless an end-of-sequence token is produced, in which case $|c| \leq L$, we define its length-normalized log-probability under model π as

$$\ell_\pi(c \mid h) = \frac{1}{|c|} \sum_{j=1}^{|c|} \log \pi(c^j \mid h, c^{<j}). \quad (1)$$

We consider two scoring functions. Likelihood-Guided Selection scores using only the large model's likelihood:

$$S_{\text{LGS}}(c_i) = \ell_{\pi_l}(c_i \mid h). \quad (2)$$

Contrastive-Guided Selection scores by subtract-

ing the small model's likelihood:

$$S_{\mathrm{CGS}}(c_i) = \ell_{\pi_l}(c_i \mid h) - \ell_{\pi_s}(c_i \mid h). \quad (3)$$

At each iteration, we append the highest-scoring chunk to the context,

$$h \leftarrow (h, c_{i^\star}), \qquad i^\star = \arg\max_{i \in \{1,\ldots,k\}} S(c_i), \quad (4)$$

where S is either S_{LGS} or S_{CGS}. We repeat until the selected chunk contains an end-of-sequence token.

4 Experiments

Our experiments evaluate whether Chunk-Level Guided Generation can improve small-model mathematical reasoning without training a reward model. We ask five questions: (i) can it match PRM guidance under the same intervention budget (Zhang et al., 2025; She et al., 2025)? (ii) why use fixed-length chunks rather than natural reasoning steps, and how sensitive are results to the choice of L? (iii) how does it compare with post-hoc selection methods such as majority voting (Wang et al., 2023), Best-of-N (Cobbe et al., 2021), and self-certainty (Kang et al., 2025)? (iv) do the gains persist when the small model is scaled from 1–1.5B to 7B parameters? and (v) do the gains from Chunk-Level Guided Generation come from generating longer reasoning traces?

4.1 Experimental Setup

Benchmarks and models. We evaluate on five mathematical reasoning benchmarks: GSM8K (Cobbe et al., 2021) (1,319 questions), MATH (Hendrycks et al., 2021) (500 questions), Minerva Math (Lewkowycz et al., 2022) (272 questions), AMC23 (40 questions), and AIME24 (30 questions). Due to the small size of AMC23 and AIME24, we run all experiments on these two datasets with three different random seeds and report averages. Our main experiments use two small-to-large model pairs: Qwen2.5-1.5B → Qwen2.5-32B (Team, 2024) and Llama-3.2-1B → Llama-3.1-70B (Meta AI, 2024). To test whether the gains persist with a stronger generator, we additionally evaluate Qwen2.5-7B → Qwen2.5-72B (Team, 2024). All models are served with vLLM (Kwon et al., 2023).

Chunk-level guidance. At each step, the small model samples $k \in \{8, 16, 32\}$ candidate chunks of L tokens at temperature 0.7. LGS selects the highest-scoring chunk under Eq. 2; CGS uses Eq. 3.

Prompting and evaluation. We prompt models using the template in Figure 3 in the Appendix and extract the prediction from the `"answer"` field. Correctness is evaluated by first normalizing the prediction and reference answer: we remove `\boxed{}` wrappers, standardize fraction notation, and strip units, currency symbols, commas, and whitespace. We then apply exact string matching, and also accept predictions that are numerically equivalent to the reference, such as `1/2` and `0.5`.

4.2 Baselines

We compare against baselines that differ in whether they use a scorer, which model provides the score, and when selection occurs.

Majority@k (Wang et al., 2023) generates k complete responses from the small model and selects the most frequent final answer.

Self-Certainty and Borda count (Kang et al., 2025) rerank the k completed responses using a distributional confidence score computed from a scorer model, without influencing generation. For a response token sequence $y = (y_1, \ldots, y_n)$ scored under model π, Self-Certainty is

$$\mathrm{SC}_\pi(y) = -\frac{1}{nV} \sum_{i=1}^{n} \sum_{j=1}^{V} \log(V \cdot p_\pi(j \mid q, y_{<i})),$$

where $n = |y|$, V is the vocabulary size, q is the input question, and $p_\pi(j \mid q, y_{<i})$ is the probability that model π assigns to vocabulary token j at position i; the score increases as the model's predicted distributions become more peaked. Borda count aggregates by final answer: the response ranked r-th by SC_π contributes weight $(k - r + 1)^p$ to its predicted answer, where k is the number of candidates and p controls how sharply top-ranked responses are weighted; the answer with the highest total weight is selected. We use p=0.5 in all experiments. Kang et al. (2025) use the generating model as the scorer (π=π_s); we additionally evaluate using the large model as the scorer (π=π_l), while keeping the small model as the generator in both cases. We refer to these as *Self-Certainty (small)* / *Borda count (small)* and *Self-Certainty (large)* / *Borda count (large)*, respectively.

Best-of-N (Cobbe et al., 2021) selects among the k completed small-model responses using the guided score in Eq. 2, with L=2048 chosen large enough that all responses finish before the limit.

Method	GSM8K			MATH			Minerva			AMC23			AIME24		
	k=8	k=16	k=32	k=8	k=16	k=32	k=8	k=16	k=32	k=8	k=16	k=32	k=8	k=16	k=32
Qwen2.5-1.5B → Qwen2.5-32B															
PRM guided search	**86.0**	89.2	92.3	**64.2**	**66.4**	**69.8**	25.0	28.3	32.4	36.7	39.2	43.3	5.6	6.7	**11.1**
LGS (ours)	85.4	87.6	90.4	60.4	62.2	68.2	31.6	33.8	36.0	35.8	39.2	43.3	**6.7**	7.8	**11.1**
CGS (ours)	85.7	**89.3**	**92.5**	**64.2**	66.2	69.6	**32.7**	**36.8**	**39.0**	**38.3**	**40.8**	**46.7**	**6.7**	**8.9**	**11.1**
Llama-3.2-1B → Llama-3.1-70B															
PRM guided search	65.4	71.8	77.3	40.2	43.4	**46.8**	11.0	13.2	12.5	18.3	17.5	20.0	0.0	2.2	5.6
LGS (ours)	69.8	77.3	81.2	37.0	37.6	41.8	12.9	**16.2**	15.4	17.5	23.3	22.5	**2.2**	3.3	4.4
CGS (ours)	**71.3**	**79.3**	**83.9**	**42.4**	**45.0**	46.2	**16.2**	15.1	**21.7**	**25.0**	**25.0**	**29.2**	**2.2**	**6.7**	**10.0**

Table 1: Accuracy (%) for the matched-budget comparison with PRM guided search. PRM guided search scores naturally delimited reasoning steps using Qwen2.5-Math-PRM-72B. LGS and CGS score fixed-length chunks using the L values chosen in Table 2, with the corresponding large model in each pair (Qwen2.5-32B for the Qwen pair, Llama-3.1-70B for the Llama pair). AMC23 and AIME24 results are averaged over three independent random seeds.

This is the full-response analogue of our method: the large model scores each completed response as a single unit, without mid-generation guidance.

PRM guided search (She et al., 2025) generates k candidate continuations for the next reasoning step, scores each with a trained PRM, and appends the highest-scoring step before continuing generation. In our experiments, we use Qwen2.5-Math-PRM-72B (Zhang et al., 2025) as the PRM.

4.3 Comparison with PRM Guided Search

We compare our methods against PRM guided search under a matched guidance budget: for each dataset we select the chunk length $L \in \{10, 20, 50, 100\}$ whose average number of CGS interventions per response most closely matches the average number of PRM scoring steps at k=8. Table 2 reports the chosen L and the resulting intervention counts for both model pairs.

Table 1 reports accuracy under the matched guidance budgets from Table 2. For the Qwen pair, CGS and PRM guided search perform comparably on GSM8K and MATH: the difference is at most 0.3 pp in either direction across $k \in \{8, 16, 32\}$. Without any reward-model training, CGS outperforms PRM guided search on Minerva Math (+7.6 pp on average across k), AMC23 (+2.2 pp), and AIME24 (+1.1 pp). LGS trails PRM on GSM8K and MATH, consistently outperforms it on Minerva Math, and is competitive on AMC23 and AIME24.

For the Llama pair, the gains are larger and more consistent. CGS outperforms PRM guided search on GSM8K, Minerva Math, AMC23, and AIME24 at every value of k, with average gains of +6.7 pp, +5.4 pp, +7.8 pp, and +3.7 pp, respectively; on MATH, CGS leads at k=8 and k=16 but trails marginally at k=32 (46.2% vs. 46.8%), with an average gain of +1.1 pp.

Dataset	Qwen 1.5B→32B			Llama 1B→70B		
	L	PRM interv.	Chunk interv.	L	PRM interv.	Chunk interv.
GSM8K	20	8.0	9.1	20	14.7	15.1
MATH	50	16.6	11.6	50	17.3	12.6
Minerva	20	23.1	26.7	20	31.5	33.9
AMC23	50	23.2	17.9	50	24.3	17.4
AIME24	50	26.5	19.3	50	25.8	25.6

Table 2: Chosen chunk length L and average number of large-model interventions per response at k=8. PRM interv. = average number of reasoning steps per response at which Qwen2.5-Math-PRM-72B scores candidates; Chunk interv. = average number of L-token chunks per response at which our method scores candidates. L is selected as the value in $\{10, 20, 50, 100\}$ whose chunk intervention count is closest to the PRM step count.

Results on AMC23 (n=40) and AIME24 (n=30) are averaged over three random seeds to reduce variance from the small test sets. Overall, chunk-level likelihood guidance from an off-the-shelf large model can match or outperform PRM guided search at a comparable intervention budget, without training or using a reward model.

4.4 Why Fixed-Length Chunks?

A natural alternative to fixed-length chunks is to guide generation at the level of reasoning steps, as in PRM guided search. However, this requires comparing candidate continuations with variable

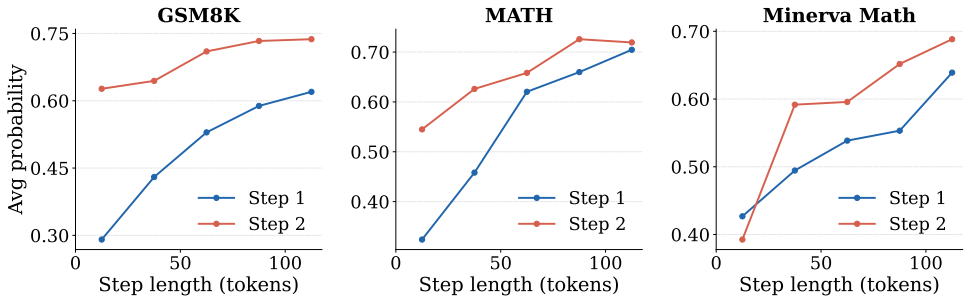


Figure 1: Length-normalized large-model probability vs. number of tokens in the first two reasoning steps for Qwen2.5-1.5B → Qwen2.5-32B, averaged over all examples in each dataset. Longer reasoning steps are assigned systematically higher probabilities by the large model even after length normalization, making variable-length step scoring unreliable for likelihood-based guidance.

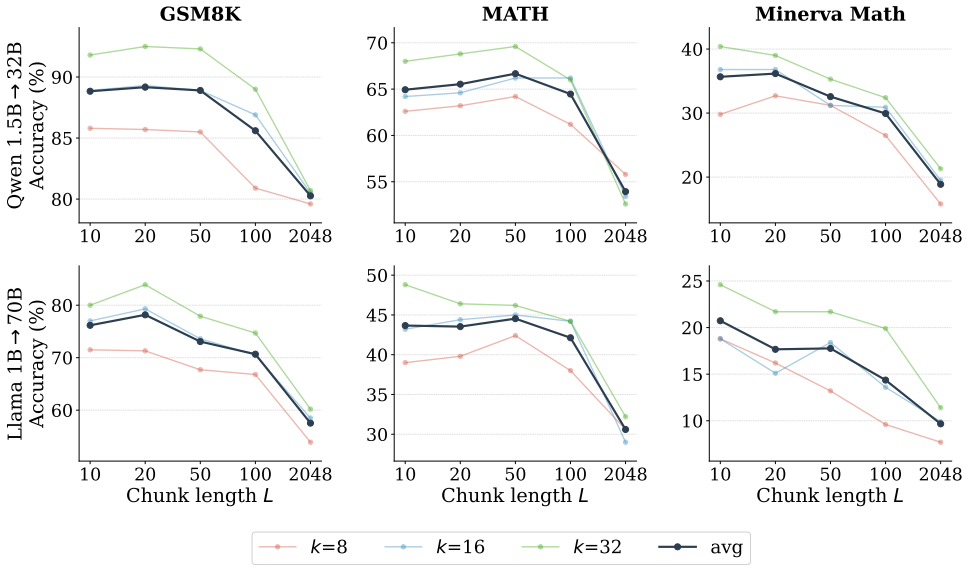


Figure 2: CGS accuracy vs. chunk length L on GSM8K, MATH, and Minerva Math for both model pairs. The rightmost point (L=2048) corresponds to Best-of-N. Performance is stable for $L \in \{10, 20\}$ across all datasets; MATH remains stable through L=50. All datasets degrade sharply at L=2048.

lengths. To test this, we run a diagnostic analysis. We collect greedy small-model responses on GSM8K, MATH, and Minerva Math and split each response into reasoning steps. For each step, we score it with the large model conditioned on all preceding steps, using the same length-normalized log-probability as in Eq. 2. We then plot the exponentiated score (average large-model probability) against step length in tokens for the first two reasoning steps, averaging over all examples in each dataset. Figure 1 shows the result for Qwen2.5-1.5B → Qwen2.5-32B.

Across datasets and for both early reasoning-step positions, longer steps receive higher scores from the large model, even after length normalization. This creates a length bias: if we used naturally delimited reasoning steps as candidates for our method, the large model would tend to prefer longer continuations, not necessarily better ones. Fixed-length chunks avoid this confound: since all k candidates at a given decision point have the same length L and are conditioned on the same context, their scores are directly comparable.

4.5 Ablation on Chunk length L

Figure 2 shows CGS accuracy as a function of L on GSM8K, MATH, and Minerva Math for both model pairs. For the Qwen pair, performance is stable for $L \in \{10, 20\}$ across all datasets; for the Llama pair, L=10 tends to perform best on Minerva Math while remaining comparable to L=20 elsewhere. On GSM8K and Minerva Math, accuracy begins to decline at L=50; MATH is less sensitive and peaks at L=50. Performance degrades further at L=100 and sharply at L=2048, where the chunk is set large enough to cover any com-

Table 3: Accuracy (%) for both model pairs; LGS and CGS use L=20. Bold = best non-oracle result per column. AMC23 and AIME24 results are averaged over three independent random seeds.

	GSM8K			MATH			Minerva			AMC23			AIME24		
Method	k=8	k=16	k=32	k=8	k=16	k=32	k=8	k=16	k=32	k=8	k=16	k=32	k=8	k=16	k=32
Qwen2.5-1.5B → Qwen2.5-32B															
Small greedy	55.6	55.6	55.6	52.8	52.8	52.8	19.1	19.1	19.1	27.5	27.5	27.5	0.0	0.0	0.0
Large greedy	94.8	94.8	94.8	78.0	78.0	78.0	50.0	50.0	50.0	52.5	52.5	52.5	16.7	16.7	16.7
Pass@k	88.6	92.7	95.1	71.0	78.6	83.8	38.2	50.7	57.4	56.7	63.3	70.0	7.8	16.7	25.6
Majority@k	74.0	77.7	79.7	56.6	59.6	63.0	21.3	24.3	26.1	28.3	36.7	34.2	2.2	3.3	4.4
Self-Certainty (small)	65.1	67.9	67.5	52.6	51.0	52.6	18.0	20.6	20.6	25.8	35.0	30.8	3.3	3.3	3.3
Borda count (small)	75.0	78.0	80.1	57.0	59.6	63.2	22.1	25.7	26.1	35.0	35.8	37.5	3.3	4.4	3.3
Self-Certainty (large)	69.7	73.4	74.8	51.2	49.2	53.6	19.9	22.4	26.5	27.5	34.2	35.8	4.4	4.4	8.9
Borda count (large)	75.2	78.7	81.0	57.6	60.0	63.4	22.8	26.8	28.3	32.5	36.7	38.3	4.4	5.6	5.6
Best-of-N	79.6	80.5	80.7	55.8	53.4	52.6	15.8	19.5	21.3	25.0	28.3	25.0	3.3	0.0	3.3
PRM guided search	**86.0**	89.2	92.3	**64.2**	**66.4**	**69.8**	25.0	28.3	32.4	36.7	39.2	43.3	5.6	6.7	11.1
LGS (ours)	85.4	87.6	90.4	60.0	61.8	67.4	31.6	33.8	36.0	35.8	40.0	40.8	3.3	**8.9**	12.2
CGS (ours)	85.7	**89.3**	**92.5**	63.2	64.6	68.8	**32.7**	**36.8**	**39.0**	**40.8**	**43.3**	**50.8**	**7.8**	**8.9**	**13.3**
Llama-3.2-1B → Llama-3.1-70B															
Small greedy	41.9	41.9	41.9	25.2	25.2	25.2	7.4	7.4	7.4	2.5	2.5	2.5	0.0	0.0	0.0
Large greedy	95.1	95.1	95.1	68.0	68.0	68.0	46.0	46.0	46.0	45.0	45.0	45.0	20.0	20.0	20.0
Pass@k	70.5	79.1	86.8	49.6	60.8	66.4	21.0	29.4	35.3	40.0	47.5	59.2	3.3	7.8	17.8
Majority@k	50.5	54.8	56.0	28.2	34.2	36.0	8.5	8.1	7.4	15.0	13.3	16.7	2.2	1.1	3.3
Self-Certainty (small)	41.6	44.7	46.6	23.0	24.4	25.8	8.8	7.0	9.2	11.7	12.5	18.3	**3.3**	0.0	1.1
Borda count (small)	51.0	55.4	57.2	28.6	33.4	37.8	8.5	9.2	9.6	14.2	16.7	18.3	2.2	1.1	2.2
Self-Certainty (large)	55.3	59.2	62.7	29.8	30.0	33.4	9.6	9.6	11.8	14.2	15.8	19.2	1.1	3.3	2.2
Borda count (large)	54.4	57.0	59.1	31.4	34.6	37.8	8.5	9.6	10.7	16.7	15.8	15.8	2.2	3.3	2.2
Best-of-N	53.9	58.5	60.2	30.6	29.0	32.2	7.7	9.9	11.4	10.8	16.7	12.5	2.2	1.1	0.0
PRM guided search	65.4	71.8	77.3	**40.2**	43.4	**46.8**	11.0	13.2	12.5	18.3	17.5	20.0	0.0	2.2	5.6
LGS (ours)	69.8	77.3	81.2	36.6	38.8	44.2	12.9	**16.2**	15.4	**24.2**	23.3	**26.7**	**3.3**	4.4	7.8
CGS (ours)	**71.3**	**79.3**	**83.9**	39.8	**44.4**	46.4	**16.2**	15.1	**21.7**	23.3	**25.0**	24.2	2.2	**10.0**	**12.2**

plete response and the large model scores only the finished output, with no opportunity to intervene during generation. We use L=20 for all remaining experiments, as it lies in the stable region across all datasets and model pairs without requiring per-dataset tuning.

4.6 Main Results

Table 3 compares all methods for the two small-model pairs (Qwen2.5-1.5B→32B and Llama-3.2-1B→70B): LGS (ours) and CGS (ours) at L=20, PRM guided search at natural reasoning-step boundaries, and post-hoc methods (Majority@k, Self-Certainty, Borda count, Best-of-N) on complete responses.

Post-hoc selection fails.

Best-of-N shows mixed results: it outperforms Majority@k on GSM8K across both model pairs at all values of k, where large-model scoring provides a reliable signal. On MATH, Best-of-N falls below Majority@k for the Qwen pair at all values of k and for the Llama pair at k=16 and k=32. On Minerva Math and AMC23, results are mixed: Best-of-N consistently underperforms Majority@k for the Qwen pair, while the Llama pair shows no clear pattern. Moreover, Best-of-N can degrade as k grows: on Qwen MATH it drops from 55.8% at k=8 to 52.6% at k=32, while Majority@k improves from 56.6% to 63.0% over the same range.

Self-Certainty (small) underperforms Majority@k in 24 out of 30 settings, showing that small-model confidence is not a reliable signal for picking the best individual response, consistent with the findings of Kang et al. (2025). Borda count (small) recovers some of this by falling back on the voting mechanism, offering marginal gains over Majority@k across most settings. Using the large model as the confidence scorer, Borda count (large) generally improves over Borda count (small), as does Self-Certainty (large) over Self-Certainty (small); Borda count (large) also outperforms Majority@k in nearly all

Method	GSM8K	MATH	Minerva	AMC23	AIME24	Avg
Small greedy	82.6	70.0	51.8	50.0	10.0	52.9
Large greedy	90.5	80.6	64.3	65.0	20.0	64.1
Pass@k	97.1	90.8	80.1	84.2	28.9	76.2
Majority@k	91.9	77.2	57.7	59.2	14.4	60.1
Self-Certainty (small)	88.2	72.8	53.3	47.5	12.2	54.8
Borda count (small)	89.8	78.0	56.6	60.0	15.6	60.0
Self-Certainty (large)	90.8	75.8	55.1	49.2	15.6	57.3
Borda count (large)	90.1	77.4	58.5	58.3	15.6	60.0
Best-of-N	92.0	75.8	48.9	50.8	11.1	55.7
PRM-72B guided search	**95.5**	**84.0**	**63.6**	65.0	**17.8**	**65.2**
LGS (ours)	92.1	80.2	62.5	60.0	16.7	62.3
CGS (ours)	91.7	81.8	**63.6**	**65.8**	15.6	63.7

Table 4: Accuracy (%) for Qwen2.5-7B → Qwen2.5-72B at k=16, L=20. Bold indicates the best non-oracle result per column. AMC23 and AIME24 results are averaged over three independent random seeds.

settings, with larger margins on the Llama pair. Post-hoc methods nonetheless remain substantially below chunk-level guidance in nearly all settings.

Pass@k as the oracle ceiling. Pass@k measures whether the correct answer appears in any of the k independent small-model samples, setting the oracle ceiling for post-hoc selection methods that choose among completed responses. On the Qwen pair at k=32, this ceiling is 95.1% on GSM8K and 57.4% on Minerva Math; no post-hoc method approaches it. Chunk-level guidance is not bound by this ceiling: by intervening during generation rather than after it, the large model can steer the small model toward solutions that no unguided sample would have reached. In three settings chunk-level guidance surpasses this ceiling: CGS on the Llama pair exceeds Pass@k on GSM8K at k=8 (71.3% vs. 70.5%) and k=16 (79.3% vs. 79.1%), and on AIME24 at k=16 (10.0% vs. 7.8%).

Chunk-level guidance substantially improves reasoning. Both LGS and CGS substantially outperform all post-hoc methods. On the Qwen pair at k=32, CGS reaches 92.5% on GSM8K and 50.8% on AMC23, gains of +12.8 and +16.6 pp over Majority@k. On the Llama pair at k=32 the gains are even larger: +27.9 pp on GSM8K (83.9%) and +14.3 pp on Minerva Math (21.7%).

Contrastive-Guided Selection outperforms Likelihood-Guided Selection. CGS outperforms LGS scoring in most settings. Rather than measuring how likely the large model finds a chunk in absolute terms, CGS measures how much more likely the large model finds it relative to the small model, capturing where the large model's guidance adds the most value. On average, CGS outperforms LGS by 2.8 pp on the Qwen pair and 2.2 pp on the Llama pair across all datasets and values of k.

Comparison with PRM at L=20. CGS outperforms PRM guided search on Minerva Math, AMC23, and AIME24 across both model pairs and all values of k. On GSM8K, CGS outperforms PRM on the Llama pair but falls marginally behind on the Qwen pair at k=8 (85.7% vs. 86.0%). MATH is the only dataset where PRM has a consistent edge: CGS falls below PRM on the Qwen pair across all k (e.g., 68.8% vs. 69.8% at k=32) and on the Llama pair at k=8 and k=32. Averaged across all five datasets and three values of k, CGS outperforms PRM by +2.8 pp on the Qwen pair and +4.7 pp on the Llama pair.

4.7 Scaling to a Larger Small Model: Qwen2.5-7B → 72B

To test whether the gains hold when the small model is already competitive, we run the same setup with Qwen2.5-7B guided by Qwen2.5-72B at k=16, L=20. Table 4 reports results across all five datasets. CGS achieves an average accuracy of 63.7%, outperforming the best post-hoc selection method by +3.6 pp on average and nearly matching the 72B greedy ceiling (64.1%), surpassing it on GSM8K (91.7% vs. 90.5%), MATH (81.8% vs. 80.6%), and AMC23 (65.8% vs. 65.0%), confirming that chunk-level guidance remains effective at this scale. On average, CGS trails PRM-72B guided search by 1.5 pp; on Minerva Math it ties PRM (63.6%) and on AMC23 it surpasses it (65.8% vs. 65.0%); the largest gap to PRM is on GSM8K (91.7% vs. 95.5%). The smaller gap between the two models in this setting — 11.2 pp between 7B and 72B greedy, compared to 27.4 pp between 1.5B and 32B (Table 3) — leaves less

room for likelihood-based scoring to differentiate among candidates, which may explain why chunk-level guidance falls short of a dedicated reward model in this setting, unlike the smaller model pairs in Table 3.

4.8 Analysis of Reasoning Trace Length

Because longer reasoning traces can improve mathematical reasoning by providing more intermediate computation, we ask whether the gains from Chunk-Level Guided Generation are explained by longer generations. Table 5 suggests the opposite. We measure reasoning trace length as the number of tokens before the final answer. We compare greedy small-model and greedy large-model generation with PRM guided search and CGS on the Qwen2.5-1.5B $\rightarrow$ Qwen2.5-32B pair; the two guided methods are evaluated at k=8, with CGS using L=20.

Dataset	Greedy small	Greedy large	PRM	CGS	PRM/ CGS
GSM8K	159	164	311	173	$\times$1.80
MATH	652	385	589	492	$\times$1.20
Minerva	651	464	773	524	$\times$1.48
AMC23	906	618	911	737	$\times$1.24
AIME24	1091	729	925	884	$\times$1.05

Table 5: Average reasoning trace length in tokens on Qwen2.5-1.5B $\rightarrow$ Qwen2.5-32B. Greedy baselines use single-sample generation, while PRM guided search and CGS are evaluated at k=8; CGS uses L=20. The final column reports the PRM/CGS trace-length ratio.

The large model's greedy responses are generally shorter than the small model's, especially on harder datasets such as MATH (385 vs. 652 tokens) and AIME24 (729 vs. 1,091 tokens). This suggests that the stronger model often reaches solutions through more direct reasoning paths. Since CGS guides the small model using the large model's likelihoods, it partially inherits this conciseness: compared with the small-model greedy baseline, CGS produces shorter traces on MATH, Minerva, AMC23, and AIME24, moving closer to the large model's reasoning length.

PRM guided search does not show the same pattern. On GSM8K and Minerva, PRM produces longer traces than the small-model greedy baseline (311 vs. 159 tokens on GSM8K, and 773 vs. 651 on Minerva). Directly comparing the two guided methods, CGS produces shorter reasoning traces than PRM guided search on all five datasets. The gap is largest on GSM8K, where PRM responses contain 80% more reasoning tokens than CGS responses (311 vs. 173 tokens). Thus, the gains from Chunk-Level Guided Generation do not come from simply generating longer reasoning traces; instead, large-model likelihood guidance appears to steer the small model toward more concise reasoning paths. This is consistent with prior findings that longer reasoning traces do not necessarily translate into better reasoning performance. (Hassid et al., 2025; Wu et al., 2025; Zhou et al., 2026; Chegini et al., 2025).

5 Conclusion

We presented Chunk-Level Guided Generation, a training-free method for improving small-model mathematical reasoning by using a large language model to score fixed-length candidate chunks. By intervening at each chunk step rather than only after generation is complete, our method can steer the small model before incorrect reasoning paths develop, addressing a core limitation of post-hoc selection. An off-the-shelf large language model can be an effective scorer. Across Qwen2.5-1.5B $\rightarrow$ 32B and Llama-3.2-1B $\rightarrow$ 70B on GSM8K, MATH, Minerva Math, AMC23, and AIME24, chunk-level guidance substantially outperforms majority voting, Best-of-N, and self-certainty-based reranking, while matching or outperforming PRM guided search on most benchmarks without reward-model training. CGS, which rewards chunks where the large model's preference diverges from the small model's, provides gains in most settings. A key design insight is the use of fixed-length chunks: large-model log-probabilities are systematically biased toward longer reasoning steps, making variable-length step scoring unreliable.

Limitations

All experiments are on mathematical reasoning benchmarks; whether chunk-level likelihood guidance generalizes to other domains such as coding or commonsense reasoning remains an open question. While our method avoids reward-model training, it requires access to a larger language model for scoring during inference, and the benefit of this guidance may diminish as the capability gap between the generator and scorer narrows. Finally, our length-bias analysis motivates fixed-length chunks, but whether a better-calibrated variable-length scoring scheme could match our approach remains unexplored.

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A Cross-Family Guidance

To examine whether the gains from chunk-level guidance extend across model families, we evaluate a cross-family setting with Llama-3.2-1B as the generator and Qwen2.5-32B as the likelihood scorer. We use k=8 and L=20, with the same prompting and evaluation setup as in the main experiments.

Method	GSM8K	MATH	Minerva
Small greedy	41.9	25.2	7.4
Majority@8	50.5	28.2	8.5
Pass@8	70.5	49.6	21.0
CGS (Llama→Qwen)	**64.4**	**38.0**	**14.0**

Table 6: Cross-family guidance with Llama-3.2-1B as the generator and Qwen2.5-32B as the scorer. CGS uses k=8 and L=20. Pass@8 is an oracle upper bound and is shown for reference.

Cross-family CGS substantially outperforms Majority@8 on all three benchmarks, with gains of 13.9, 9.8, and 5.5 percentage points on GSM8K, MATH, and Minerva Math, respectively. These results suggest that chunk-level guidance can remain effective when the generator and scorer belong to different model families. We evaluate only one cross-family direction, however, and leave a broader evaluation across model-family combinations to future work.

B Large-Model Inference Cost

We compare the large-model wall-clock cost of CGS with directly generating using Qwen2.5-32B on 100 GSM8K questions (Qwen2.5-1.5B→Qwen2.5-32B, k=8, L=20).

Method	Total time	Time/question
Large greedy	175 s	1751 ms
CGS scoring	64 s	637 ms

Table 7: Large-model wall-clock cost on 100 GSM8K questions.

Under this setup, the large-model wall-clock time for CGS is approximately $2.7\times$ lower than for direct large-model generation. This comparison measures only the large model's computation.

C Prompt Template

Solve this math problem step by step, then provide your answer.
`[PROBLEM]`

Think through the problem step by step, showing your reasoning. Then provide your final answer in this exact format:

```
{
  "reasoning": "your step-by-step solution here",
  "answer": "your_final_answer"
}
```

Put only the final answer in the `"answer"` field. Use LaTeX formatting if needed (e.g., `\frac{a}{b}` for fractions).

Figure 3: Instruction template used for all datasets. The model is prompted to reason step by step (Wei et al., 2022) and produce a structured JSON response; we extract the predicted answer from the `"answer"` field.